Lectures 3

adapted on
Lynford L. Goddard
Brian T. Cunningham

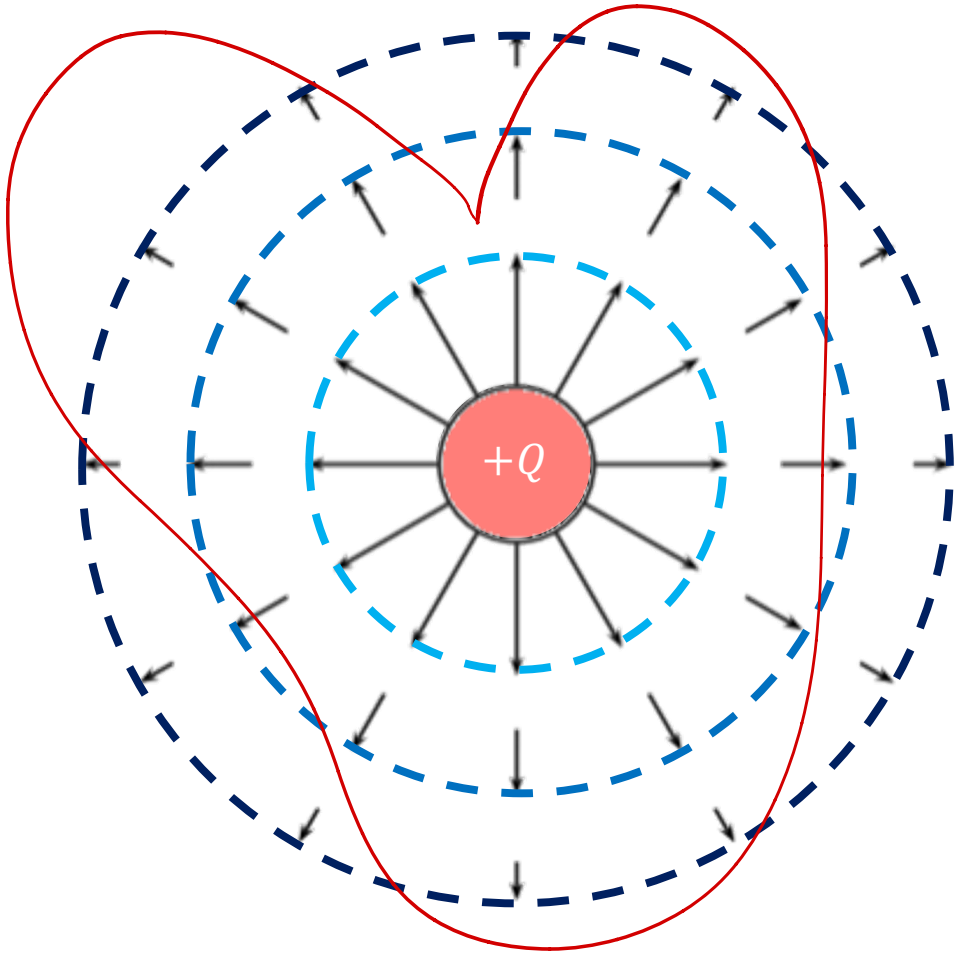
Surface Integrals
Connecting Coulomb's and Gauss' Law

$$\vec{F} = \frac{Q_1 Q_2}{4\pi\varepsilon_0 R^2}\hat{a}_R \rightarrow \vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\hat{a}_R$$

Victoria Shao
yangshao@Illinois.edu

© Lynford Goddard and Brian Cunningham - Copying, publishing or distributing without express written permission is prohibited.

Electric Field Around a Point Charge



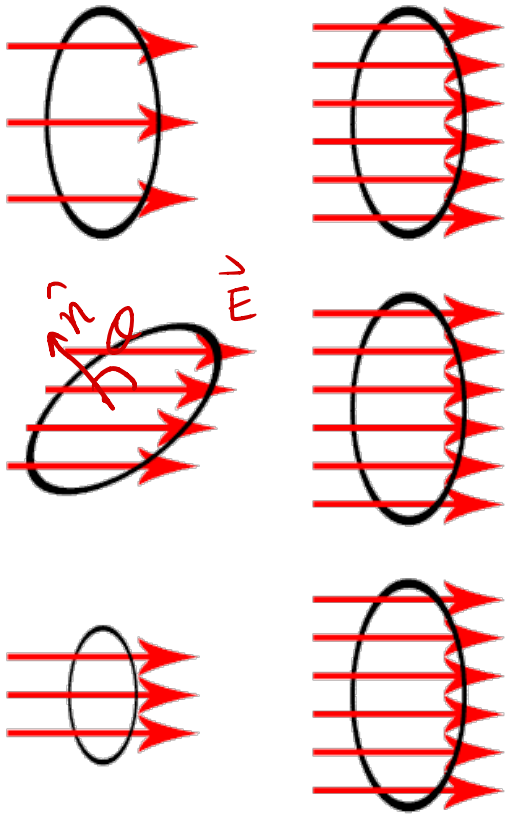
$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\,\hat{a}_R$$

Gauss' Law

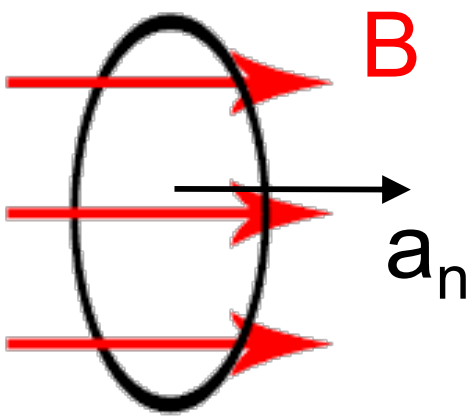
Number of field lines passing through **any** surface that encloses Q_1 is constant.

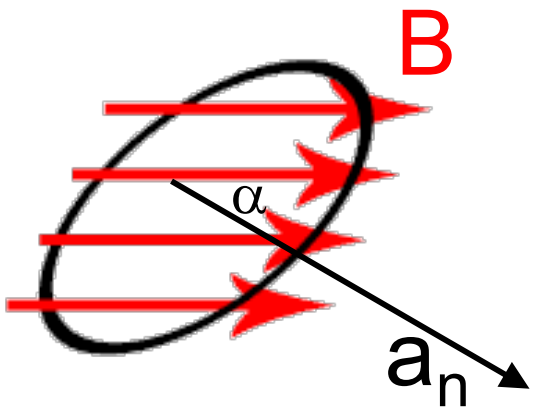
Surface Integrals

- Flux = # of arrows that pass thru a surface; it depends on:
  - The density of vectors
  - The angle of the surface
  - The area of the surface



Surface Integral describes the Flux of a Vector Field





$$Flux = \vec{B} \bullet \Delta\vec{S}$$

$$= (\vec{B} \bullet \hat{a}_n)\left|\Delta S\right|$$

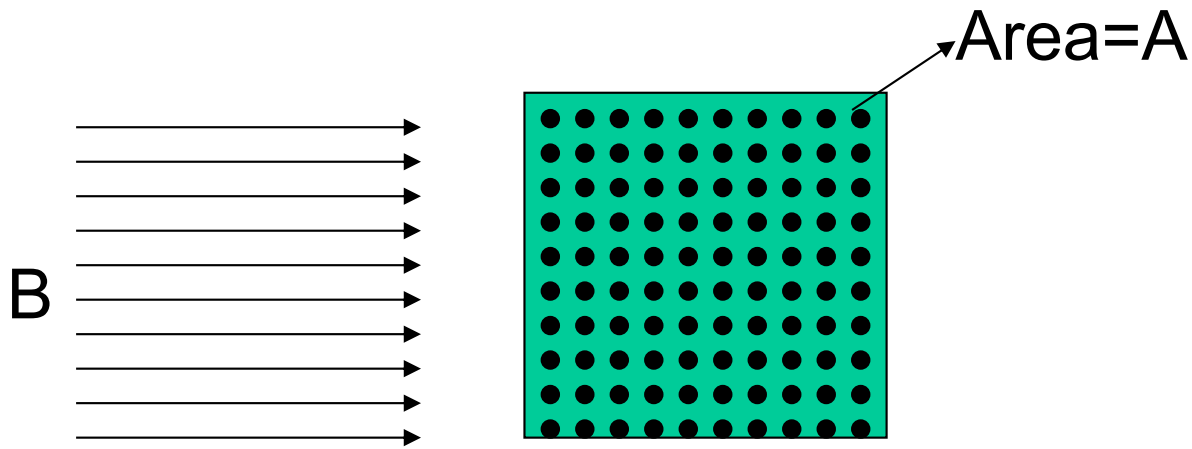
$$= (B\cos\alpha)\left|\Delta S\right|$$

↓ projection

$$= \left|B_n\right|\left|\Delta S\right|$$

Trick that works sometimes

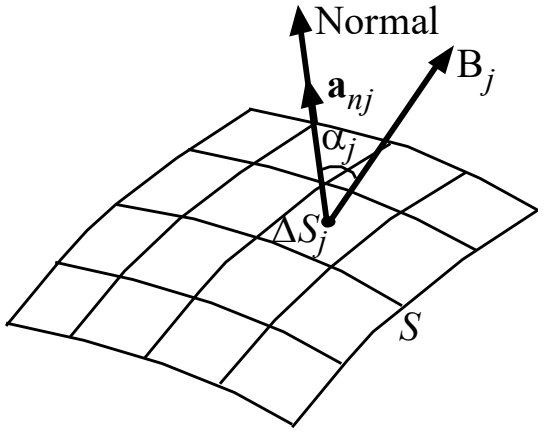
If the flux is
- Uniform (has equal magnitude across whole surface)
- Perpendicular to the surface



$$\psi = \iint_S \vec{B} \bullet d\vec{S} = B \cdot A$$

scaler.

In the general case …



$$\text{Flux} = \sum_{j=1}^{n} \Delta\psi_j$$

$$= \sum_{j=1}^{n} \mathbf{B}_j \bullet \Delta\mathbf{S}_j$$

In the limit $n \to \infty$,

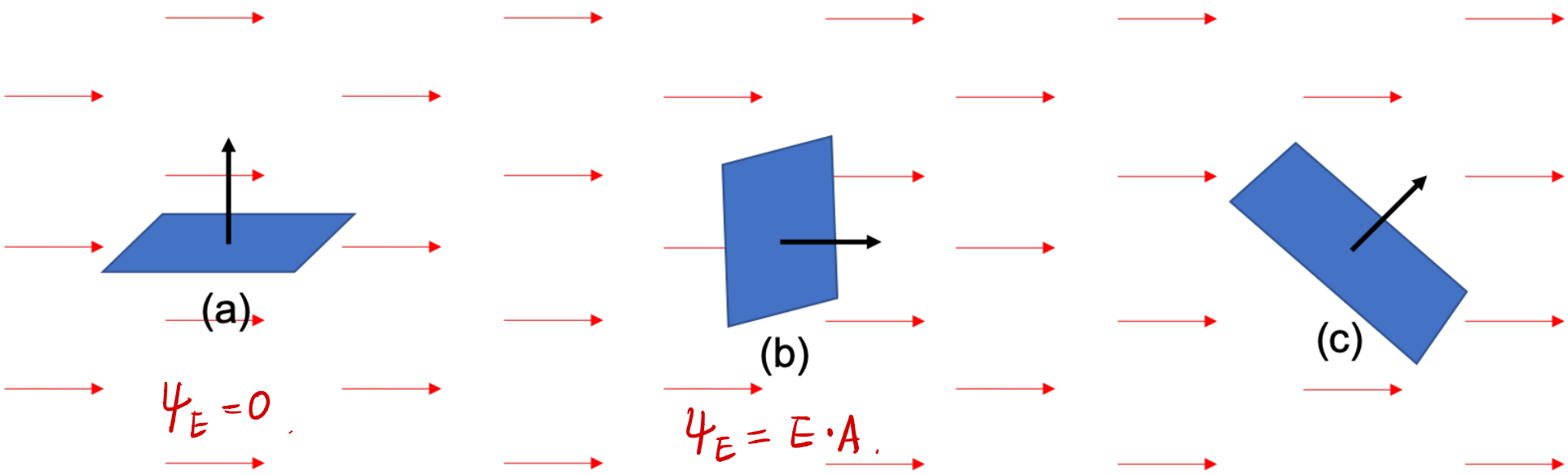
$$\text{Flux, } \boxed{\psi = \int_S \mathbf{B} \bullet d\mathbf{S}}$$

= Surface integral of B over S.

Q:

Which of the following three case has more flux?

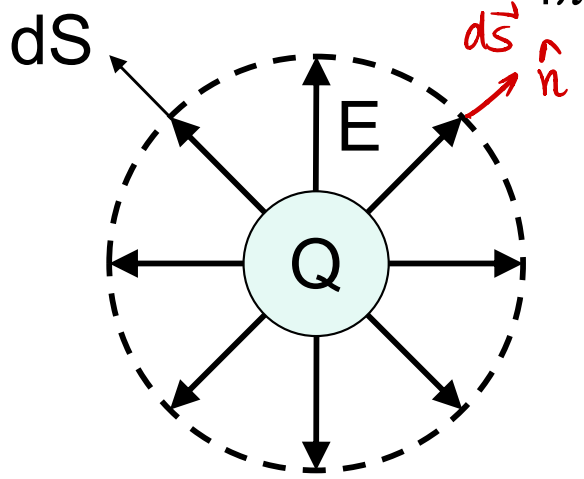
Compare $\vec{E} \cdot \vec{A}$



Electric Flux = Charge Enclosed

- Coulomb's Law for the Electric field of a point charge:

$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\hat{a}_R$$



Define **D=ε_0E** to be the "displacement flux density"

[c]

$$\psi_E = \oiint_S \vec{D} \bullet d\vec{S} = \oiint_S \varepsilon_0 \vec{E} \bullet d\vec{S}$$

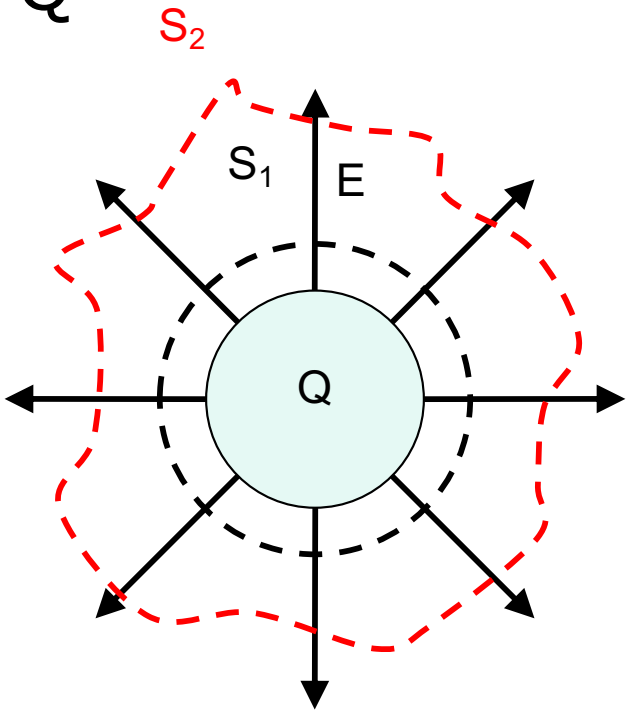
E const in mag. $\vec{E}$ and $d\vec{S}$ same direction

$$= \varepsilon_0 E(Surf\ Area)$$

$$= \varepsilon_0 \frac{Q}{4\pi\varepsilon_0 R^2}(4\pi R^2) = Q$$

Same Flux Out of Any Surface

- Same # of field lines pass thru any surface that encloses Q

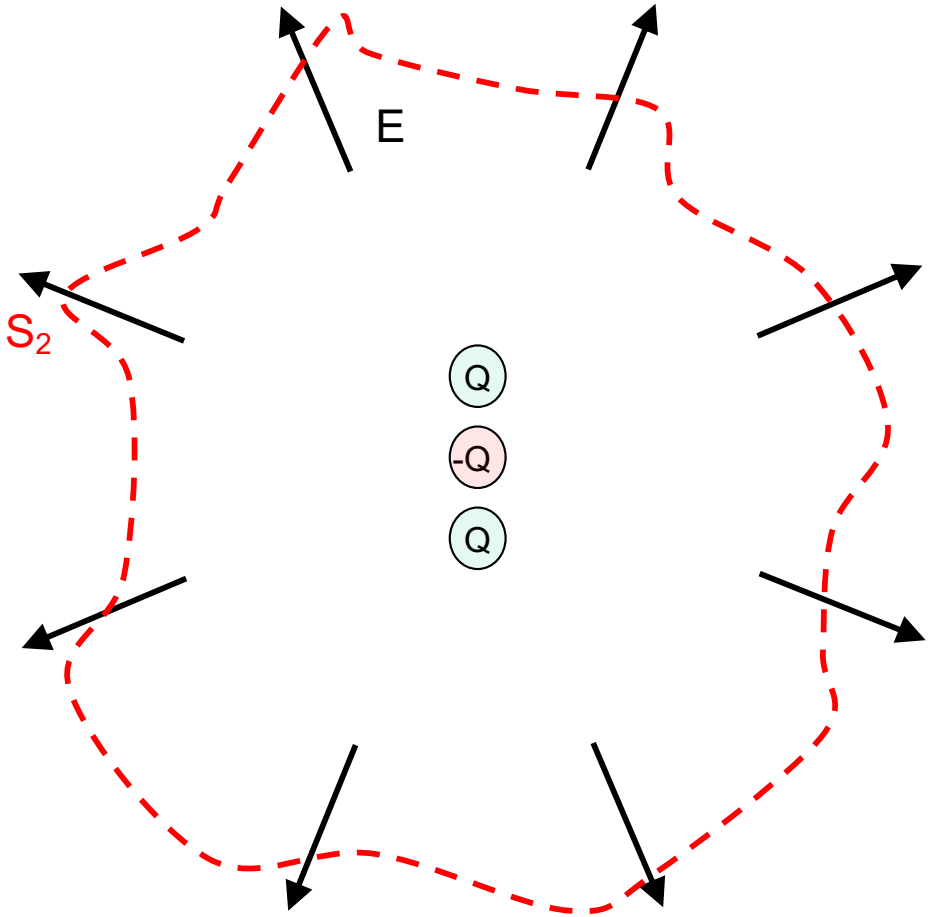


$$\psi_E = \oiint_S \vec{D} \bullet d\vec{S} = Q_{enclosed}$$

Our first Maxwell Equation!

Superposition

- The flux for an arbitrary distribution of charges is obtained using superposition



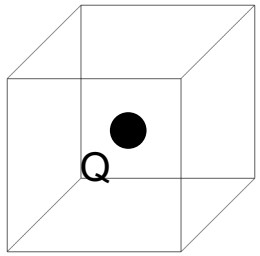
$$\psi_E = \oiint_S \vec{D} \bullet d\vec{S} = Q_{enclosed}$$

Q+Q−Q

FLUX OUT = CHARGE ENCLOSED

Simple Example 1

6-sided cube with Q at the center:

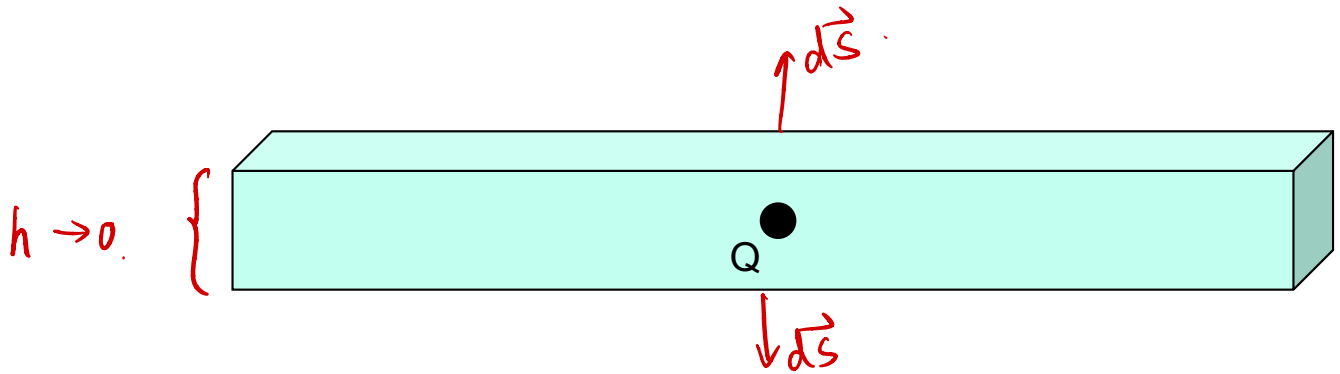


Flux out of entire box = Q
Flux out of one side = Q/6

Simple Example 2

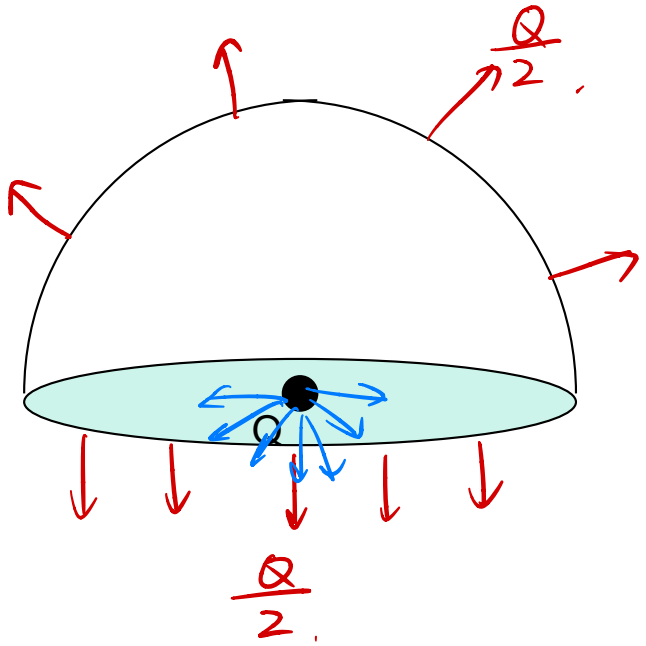
$\psi_{Bottom} = \frac{Q}{2}$

Flux pointing out of “thin” box on bottom surface with Q at the center



Simple Example 3

Flux out of hemisphere with Q at the center =

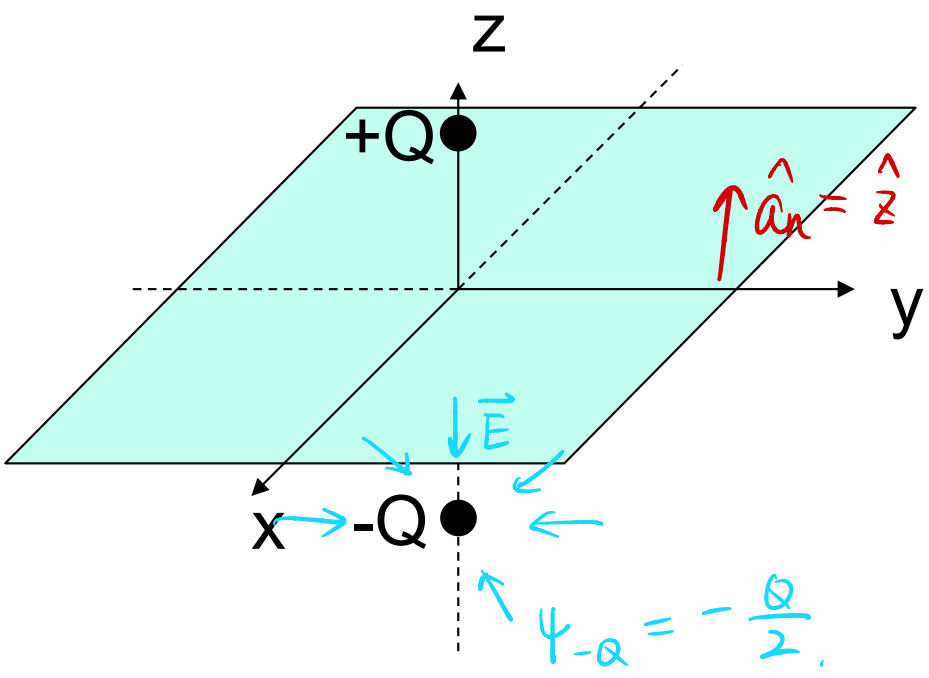


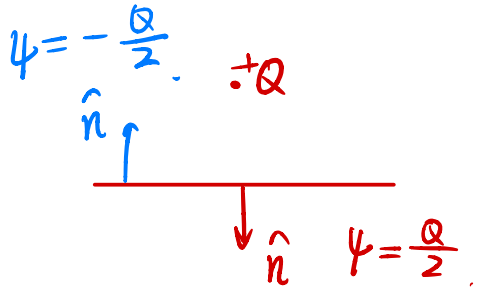
Challenge Question 1

- What is the flux across the xy plane $\mathbf{a}_n=\mathbf{a}_z$ for a dipole: +Q at (0,0,a/2) and -Q at (0,0,-a/2)? (Hint: Use superposition.)

(a) Q/2
(b) -Q/2
(c) -Q
(d) -3Q/2
(e) -2Q

$-\frac{Q}{2} - \frac{Q}{2}$.

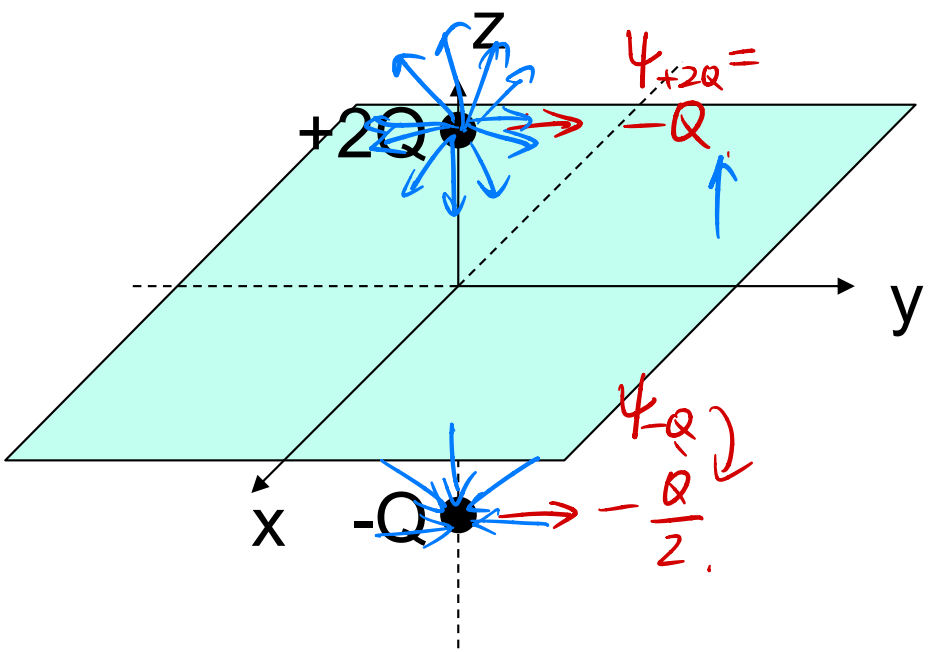




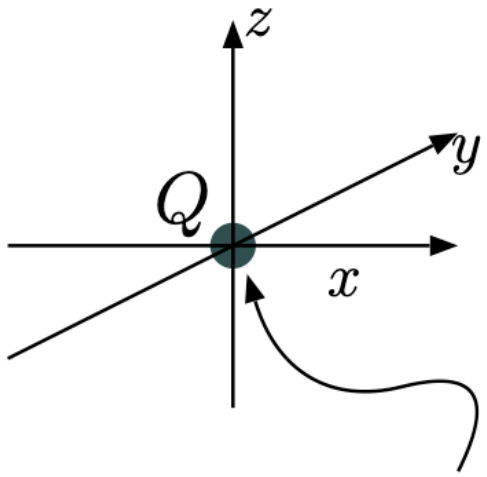
Challenge Question 2

- What is the flux across the xy plane $\mathbf{a}_n=\mathbf{a}_z$ for the charge distribution: +2Q at (0,0,a/2) and -Q at (0,0,-a/2)? (Hint: Use superposition.)

(a) Q/2
(b) -Q/2
(c) -Q
(d) -3Q/2
(e) -2Q



Charge use Impulse Notation



$[C/m^3]$ volumetric

Q: point charge. $[C]$

$$\rho(x,y,z) = Q\delta(x)\delta(y)\delta(z)$$ → $[\frac{1}{m}]$

From ECE 210 $\int_{-\infty}^{\infty} \delta(t)dt = 1$

$\int_{-\infty}^{\infty} \delta(x)dx = 1$

$[\frac{1}{m}] \cdot [m] = 1$

isolated point charges at $\vec{r} = (x_0, y_0, z_0)$

$$\rho(x,y,z) = Q\delta(x-x_0)\delta(y-y_0)\delta(z-z_0) \quad \text{C/m}^3$$

$$\int_V \rho dV = \iiint Q\delta(x-x_0)\delta(y-y_0)\delta(z-z_0)dxdydz = Q$$

Charge use Impulse Notation

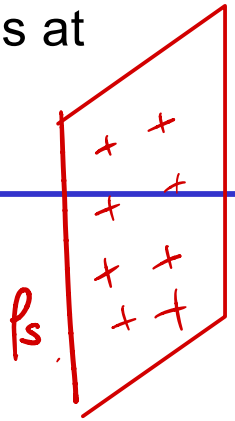
line charges at
$x = x_0$ and $y = y_0$



$$\rho(x, y, z) = \lambda\delta(x - x_0)(y - y_0)$$

ρ_ℓ:
λ: [C/m]

surface charges at
$x = x_0$



$$\rho(x, y, z) = \rho_s\delta(x - x_0)$$

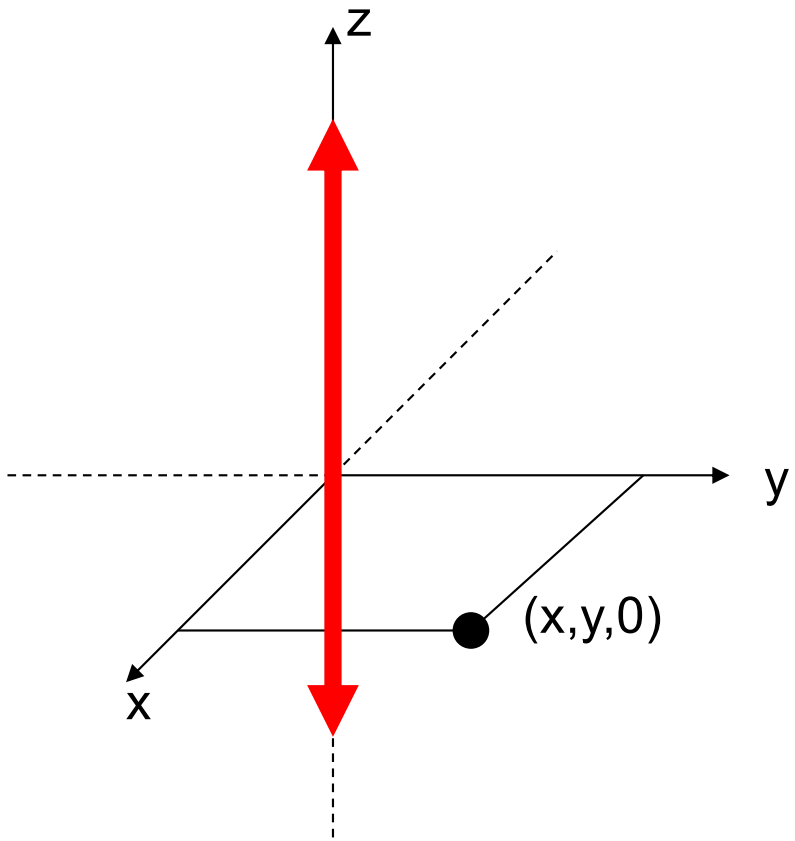
ρ_s : [C/m^2]

$$Q = \int_\ell \rho_\ell \, d\ell$$

$$= \int_S \rho_s \, ds$$

$$= \int_V \rho \, dv$$

Example 3: **E** due to ∞ line of charge

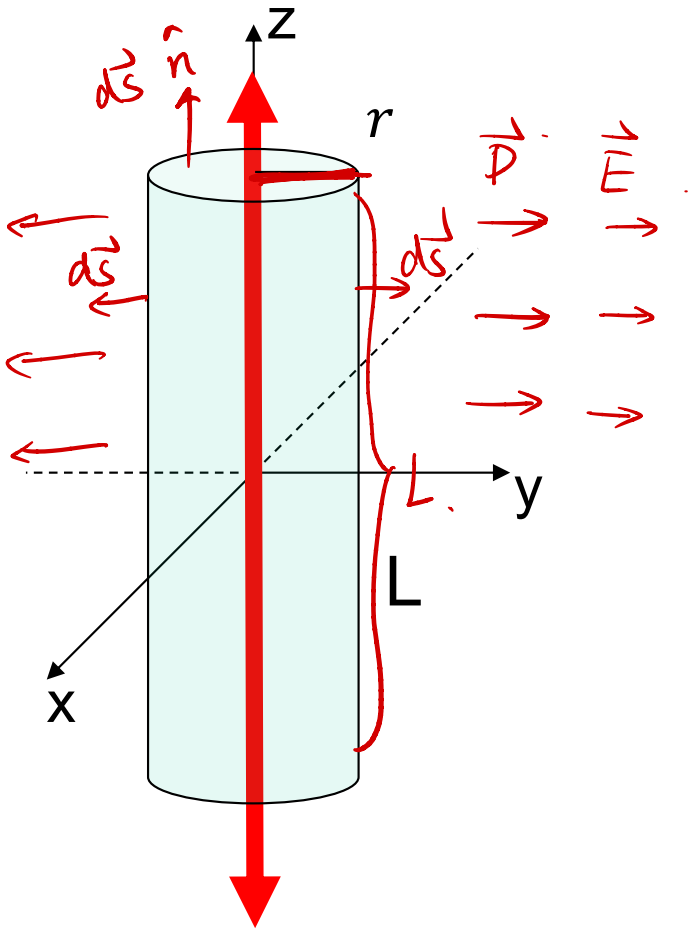


Linear charge distribution ρ_l (C/m)

Find **E** at point on xy plane

$$\oint_S \vec{D} \cdot d\vec{s} = Q$$

Example 3: **E** due to ∞ line of charge



Gaussian pillbox

$[D] \cdot m^2 \quad [C]$

$$\oint_s \vec{D} \cdot d\vec{s} = Q$$

$D : [C/m^2]$

Top + Bottom + Side

0 + 0 + $D \cdot \text{Area} = \int_V \rho(x,y,z)\, dv$

$= \int \rho_l dl$

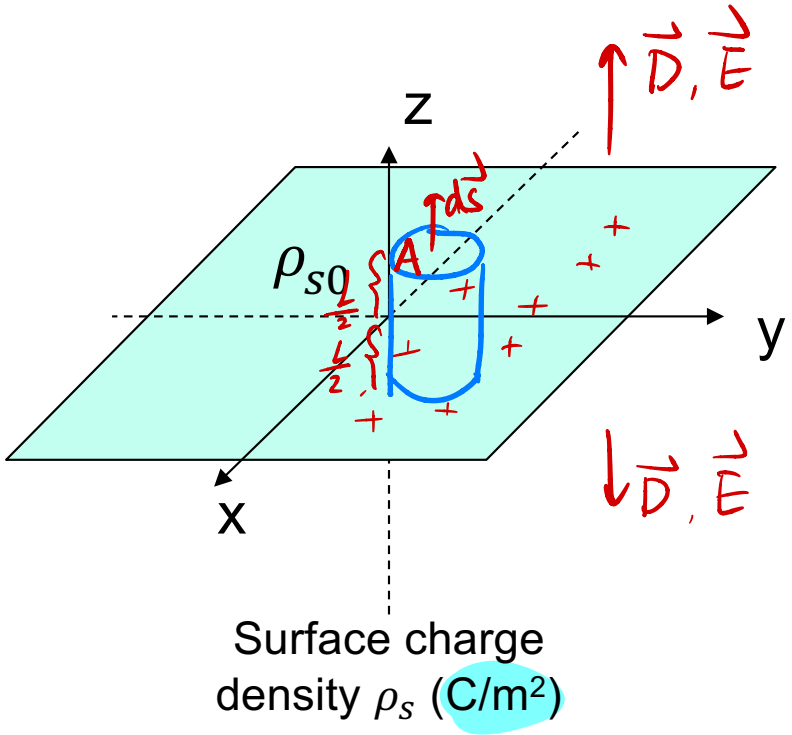
$$D \cdot 2\pi r L = \rho_l L$$

$$\vec{D} = \frac{\rho_L}{2\pi r} \hat{r} \quad [C/m^2]$$

$$\vec{E} = \frac{\vec{D}}{\varepsilon_0} = \frac{\rho_l}{2\pi\varepsilon_0 r} \hat{r} \quad [V/m]$$

↑ Same result of Lect 2.

Example 4: **E** due to a surface of charge



Point charges Q are distributed over z = 0 plane with an average surface charge density of ρ_{s0} C/m². Find **E**.

$$\oint\oint \vec{D}\cdot d\vec{s} = Q$$

Top + Bottom + Side

$$D\cdot A + D\cdot A + 0 = \iint_S \rho_s \cdot ds$$

$$2DA = \rho_s \cdot A$$

$$\vec{D} = \frac{\rho_s}{2}\,\mathrm{sgn}(z)\ [C/m^2]$$

$$\mathrm{sgn}(z) = \begin{cases} 1 & z>0 \\ -1 & z<0 \end{cases}$$

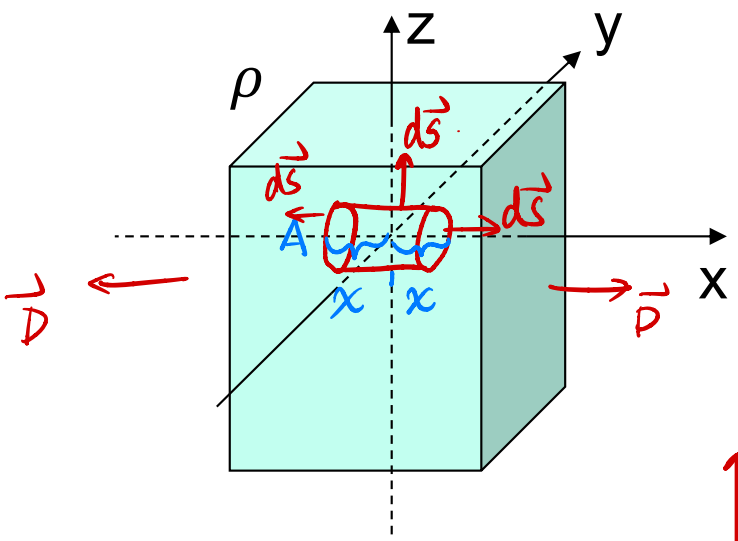
$$\vec{E} = \frac{\vec{D}}{\varepsilon_0} = \frac{\rho_s}{2\varepsilon_0}\,\mathrm{sgn}(z)$$

[V/m]

(Review at home)

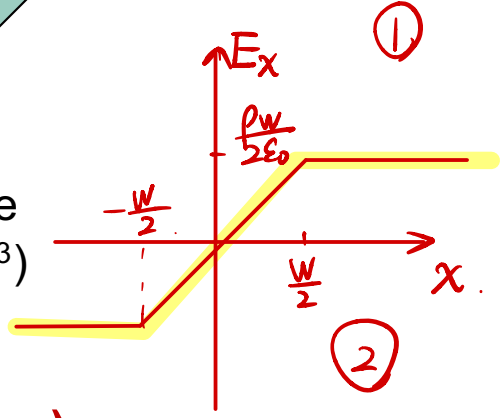
Prof. Kudeki's lect03.pdf
Example 1

Example 5: **E** due to a slab of charge



Volume charge density ρ (C/m^3)

Point charges Q are distributed throughout an infinite slab of width W located over $-\frac{W}{2} < x < \frac{W}{2}$. Find **E**.



(Review at home)

Prof. Kudeki's lect03.pdf
Example 2

$$\oiint \vec{D} \cdot d\vec{s} = Q = \iiint_V \rho \, dV$$

Left + Right + Side

$D \cdot A + D \cdot A + 0 = \rho \cdot A \cdot 2x$

$\vec{D} = \rho x \, \hat{x} \quad |x| < \frac{W}{2}$ $\quad |x| < \frac{W}{2}$

$D \cdot A + D \cdot A + 0 = \rho \cdot A \cdot W$

$|x| \geq \frac{W}{2}$

$\vec{D} = \frac{\rho W}{2} \hat{x} \, \mathrm{sgn}(x) \quad |x| \geq \frac{W}{2}$

$$\vec{E} = \frac{\vec{D}}{\varepsilon_0} = \begin{cases} \frac{\rho x}{\varepsilon_0} \hat{x} & |x| < \frac{W}{2} \\ \frac{\rho W}{2\varepsilon_0} \hat{x} \, \mathrm{sgn}(x) & \text{else} \end{cases}$$

Summary of electrostatic fields E

- Coulomb field

$$\mathbf{E} = \hat{r}\frac{Q}{4\pi\epsilon_o r^2} \text{ of a point charge } Q,$$

- Field

$$\mathbf{E} = \hat{r}\frac{\lambda}{2\pi\epsilon_o r} \text{ of constant line density } \lambda,$$

- Field

no distance

$$\mathbf{E} = \hat{x}\frac{\rho_s}{2\epsilon_o}\text{sgn}(x) \text{ of constant surface density } \rho_s,$$

$\vec{D} = \epsilon_0\vec{E} = \frac{\rho_s}{2}\hat{x}\,\text{sgn}(x)$

- Field

$$\mathbf{E} = \hat{x}\frac{\rho x}{\epsilon_o} \text{ of constant volume density } \rho$$

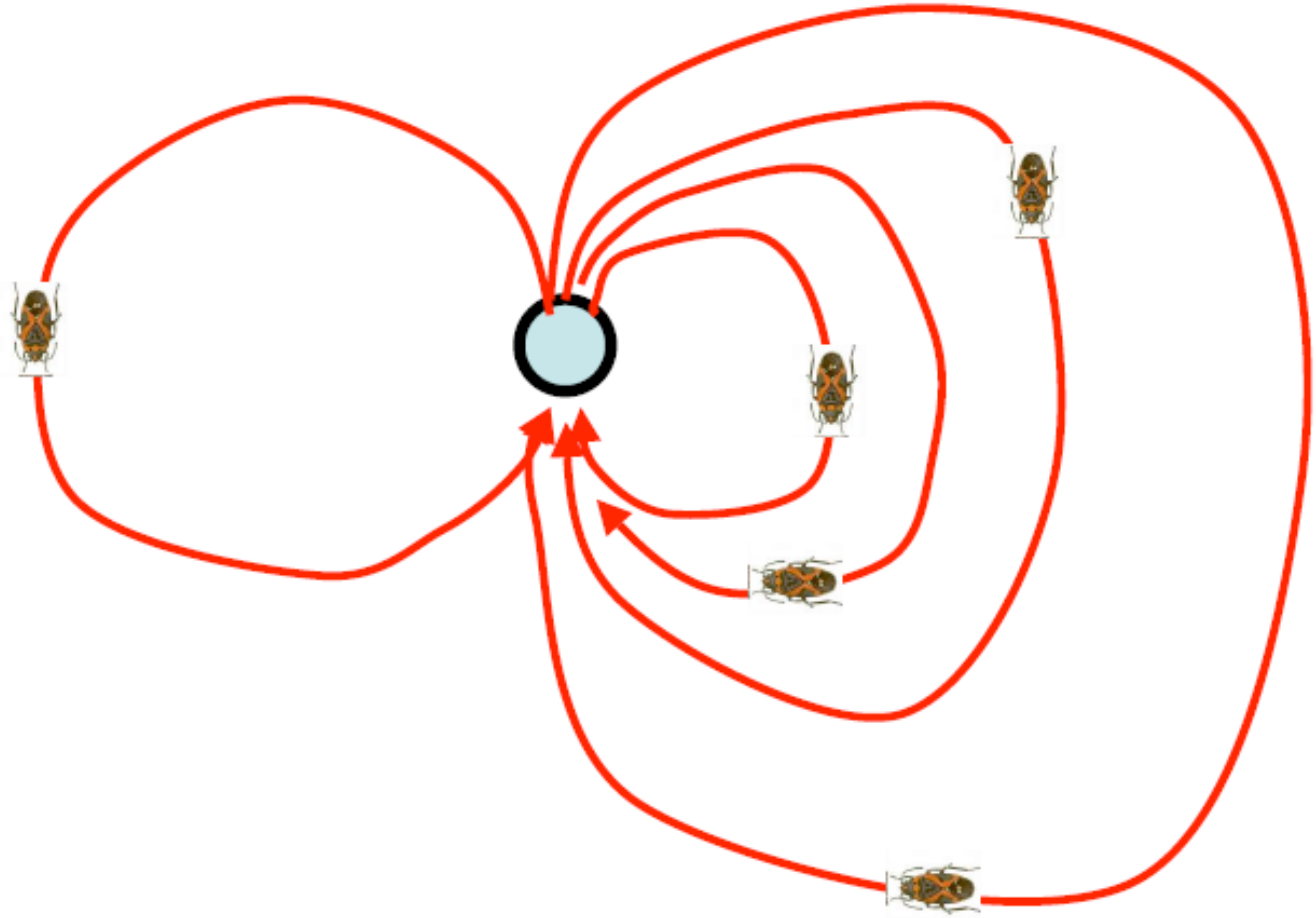
$|x| < \frac{W}{2}$

The (Fictional) Yadaraf Bug: Flux and Surface Integrals



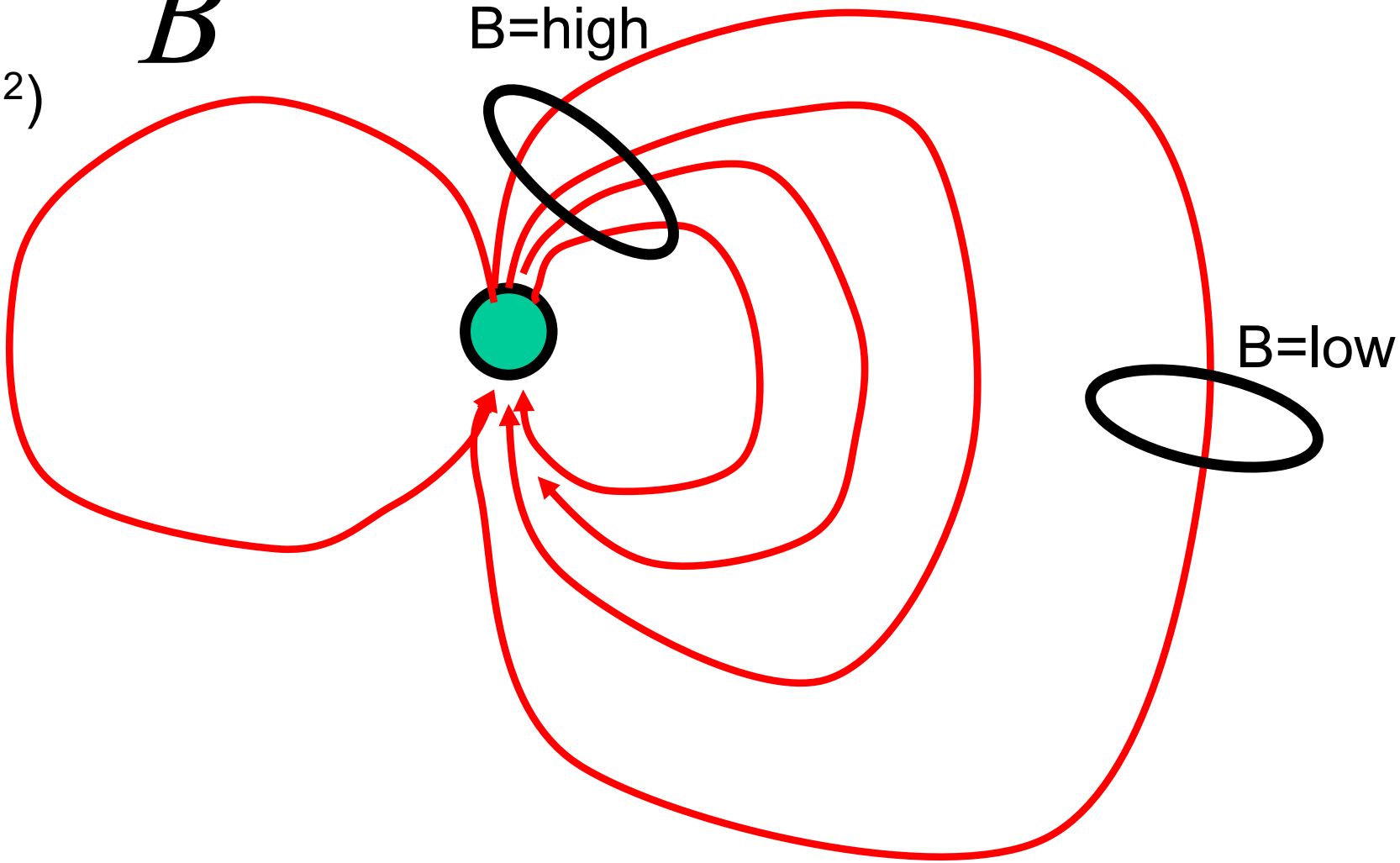
- The Yadaraf Bug
  - They live in the ground
  - They only come out at night to search for food
  - Very hard to see - REALLY small
  - After gathering food, they <u>always</u> return to their hole in the ground

Yadaraf Bug Travel Paths

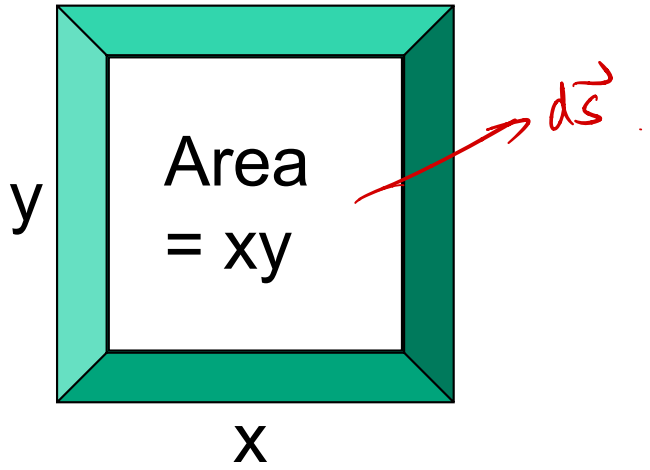


Yadaraf Bug Travel Paths

Bug Density Vector (Bugs/m^2) $\vec{B}$



Yadaraf Bug Counter

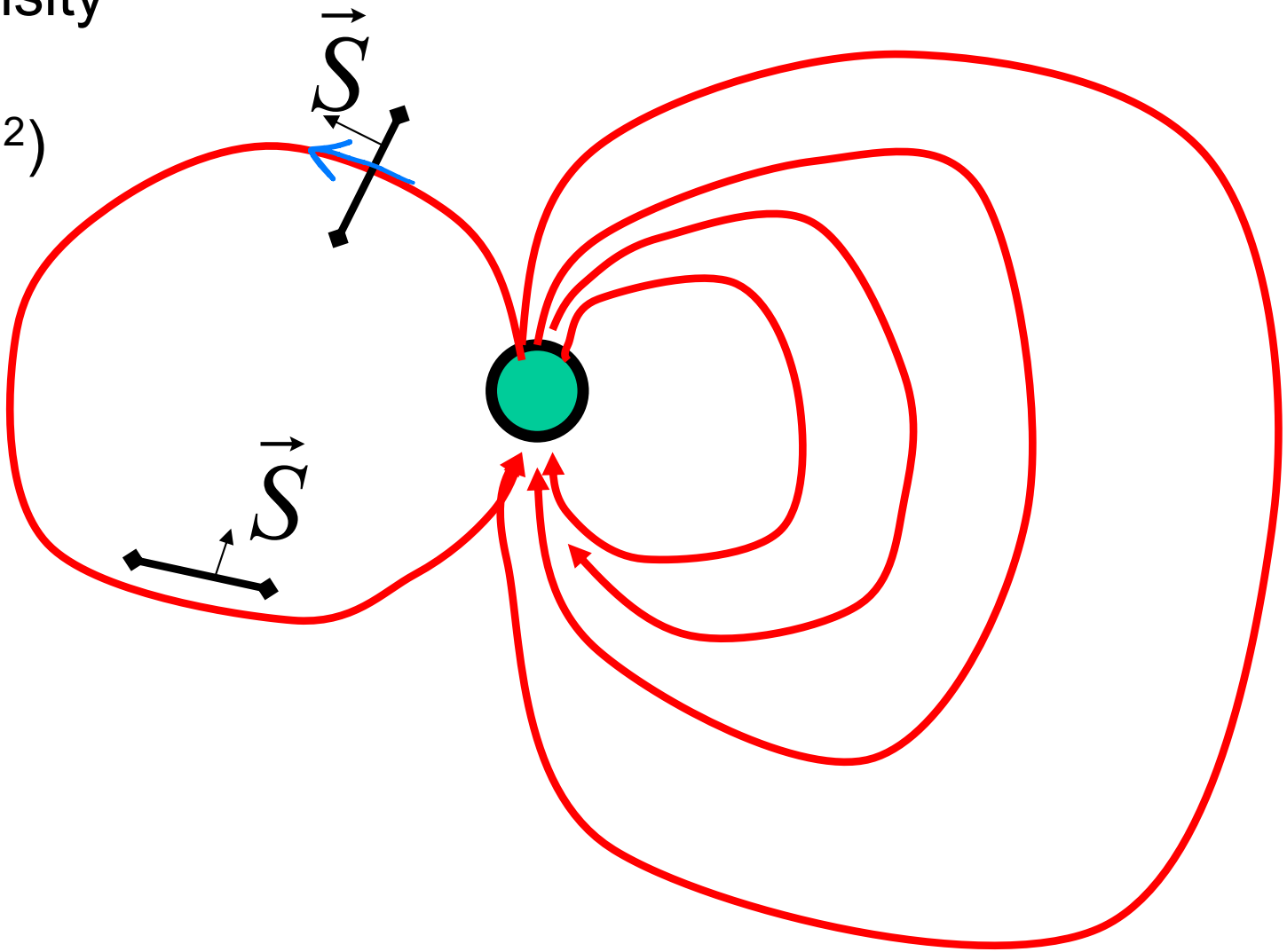


- Register a +1 count for each bug going through in one direction
- Register a -1 cound for each bug going through in the opposite direction
- Has a known area for the bugs to pass through

Yadaraf Bug Travel Paths

Bug Density Vectors (Bugs/m^2)

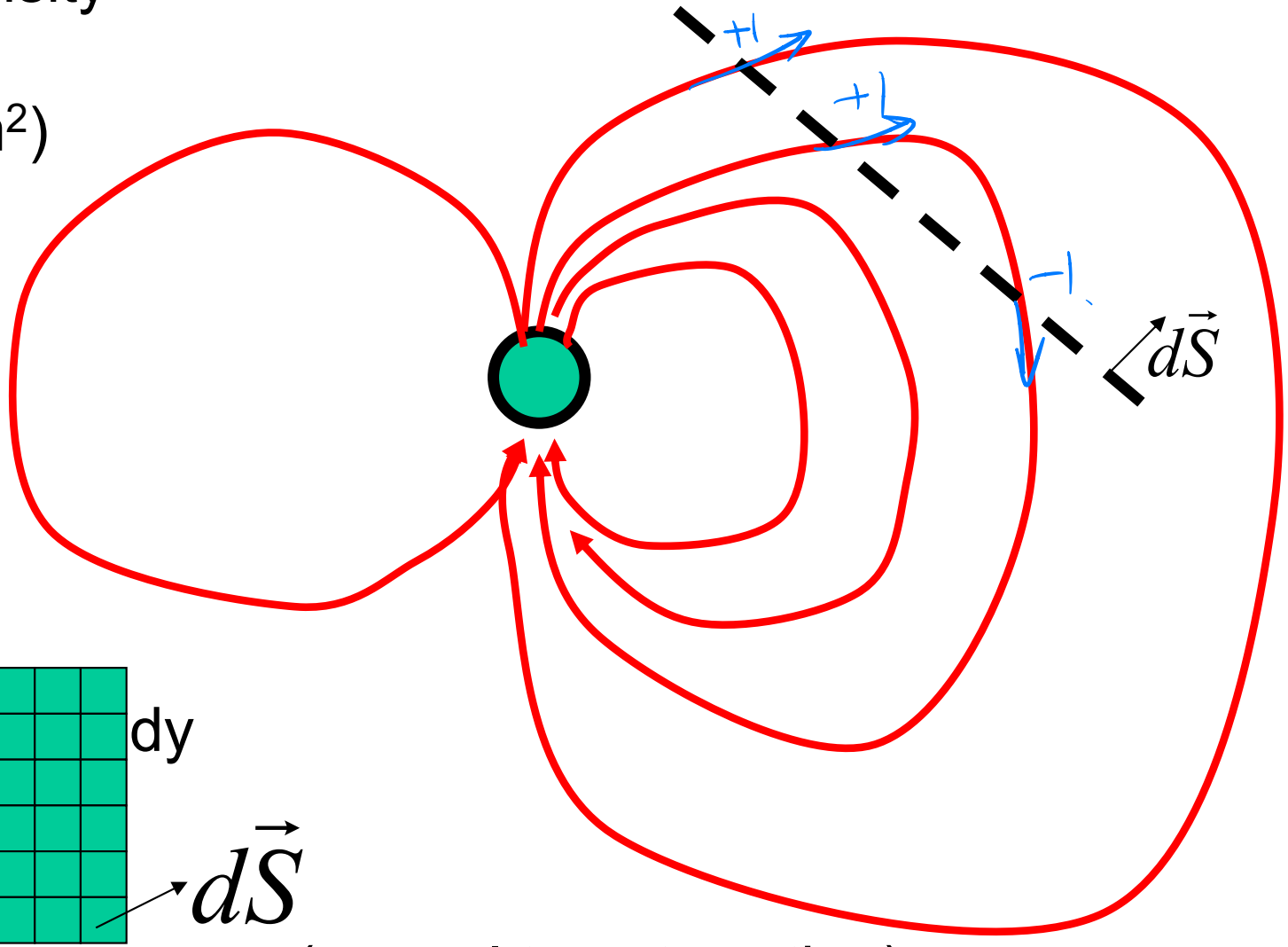
$\vec{B}$

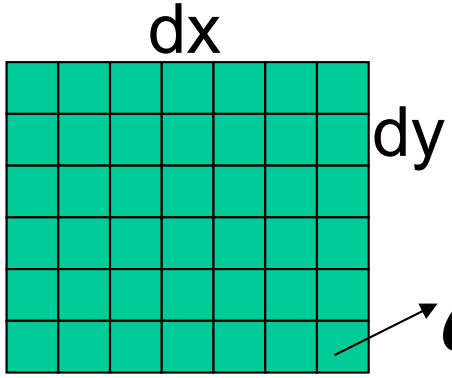


Bug Counting Net

Bug Density Vectors (Bugs/m^2)

$\vec{B}$



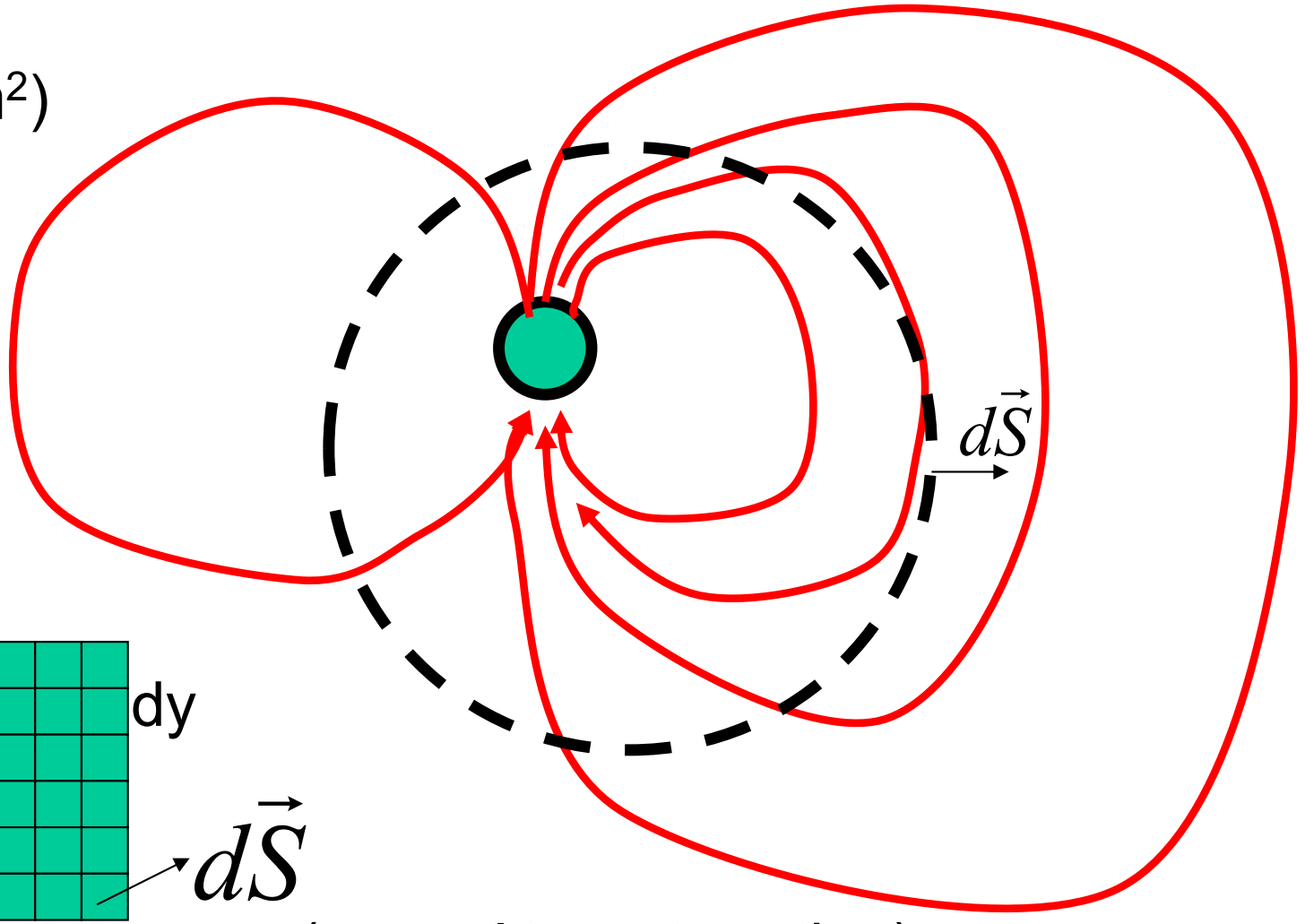


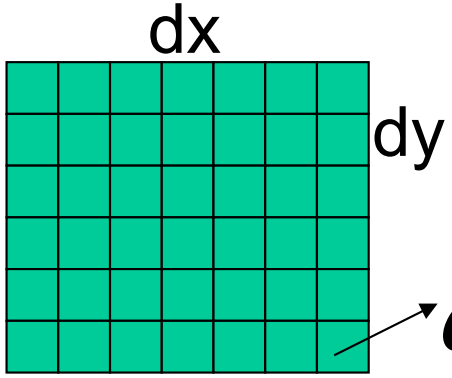
(normal to net section)

"Closed" Bug Counting Net

Bug Density Vectors (Bugs/m^2)

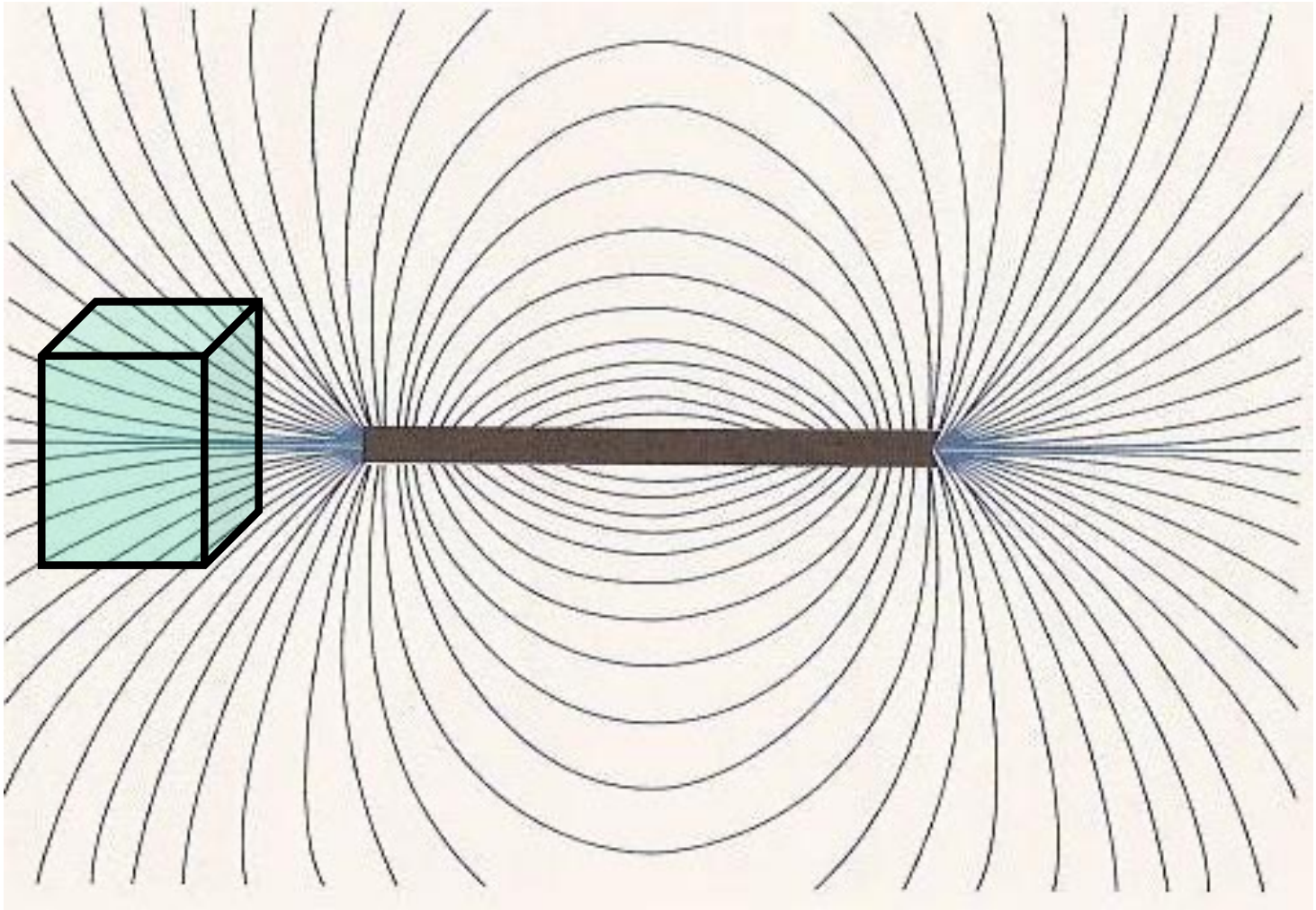
$\vec{B}$





(normal to net section)

So magnetic flux, ψ_B, through a CLOSED surface is always zero



Magnetic field lines form closed paths (they do not begin or end)

Gauss' Law for B Fields

Net flux of magnetic field lines through any closed surface MUST be zero.

$$\oiint_S \vec{B} \bullet d\vec{S} = 0$$

Our Second Maxwell Equation!

Lecture 3 Summary

- Gauss' Law

$$\psi_M = \oiint \vec{B} \cdot d\vec{S} = 0$$

$$\psi_E = \oiint_S \vec{D} \bullet d\vec{S} = Q_{enclosed}$$

FLUX OUT = CHARGE ENCLOSED

- Review examples on sheet charges, slab charges (slides 23-26)

- Next lecture
  - Divergence and curl